

An exponential function is of the form

$$f(x) = b^x \text{ for some } b \in \mathbb{R}.$$

If $x \in \mathbb{Q}$, this makes sense as $x = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ then

$$b^{\frac{p}{q}} = \sqrt[q]{b^p} = \left(\sqrt[q]{b}\right)^p$$

but if x is irrational, this makes less sense. It turns out for any irrational number, there is a sequence of rational numbers that converge to it. So we define, for x irrational,

$$b^x = \lim_{y \rightarrow x} b^y \text{ where } y \text{ is rational.}$$

this makes b^x a continuous smooth function on all real numbers. So we can define

$$b^\pi \text{ as } \lim_{y \rightarrow \pi} b^y, \text{ with } y = 3, 3.1, 3.14 \dots$$

we'll find $\frac{d}{dx}(b^x)$ later.

Some Exponent Rules

If $b > 0$, $b \neq 1$, $f(x) = b^x$ is increasing with domain $\mathbb{R}$ and range $(0, \infty)$. If $0 < b < 1$ then $f(x) = b^x$ is decreasing.

Take $a, b > 0$, $x, y \in \mathbb{R}$

1. $b^{x+y} = b^x b^y$

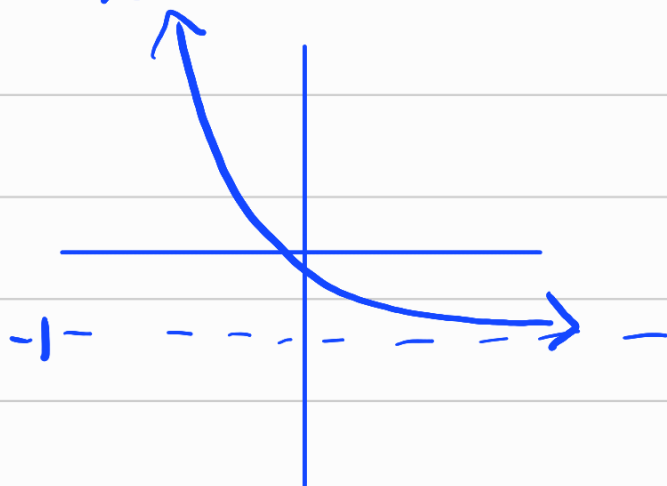
2. $b^{x-y} = b^x b^{-y} = \frac{b^x}{b^y}$

3. $(b^x)^y = b^{xy}$

4. $(ab)^x = a^x b^x$

$$\text{If } 0 < b < 1, \quad \lim_{x \rightarrow \infty} b^x = 0, \quad \lim_{x \rightarrow -\infty} b^x = \infty$$
$$b > 1 \quad \lim_{x \rightarrow \infty} b^x = \infty, \quad \lim_{x \rightarrow -\infty} b^x = 0$$

Ex $\lim_{x \rightarrow \infty} 2^{-x} - 1 = \lim_{x \rightarrow \infty} \frac{1}{2}^x - 1 = 0 - 1 = -1$



$$\frac{d}{dx}(b^x) = ?$$

$$\text{let } f(x) = b^x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h}$$

$$= \lim_{h \rightarrow 0} b^x \left(\frac{b^h - 1}{h} \right)$$

$$= b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h}$$

$$= b^x f'(0)$$

So, $f'(x)$ exists if $f'(0)$ exists.

We'll discuss later, but $f'(0) = \ln(b)$

Let's talk about e .

Def, e is a real, irrational, number
s.t.

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

* historically, $e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

Then we got a wonderful, and extremely versatile, relation

$$\frac{d}{dx}(e^x) = e^x, \quad \int e^x dx = e^x + C$$

Ex $y = e^{\tan x}$

$$y' = \sec^2 x e^{\tan x}$$

In general, $\frac{d}{dx}(e^{f(x)}) = f'(x)e^{f(x)}$

Ex $\int x^2 e^{x^3} dx, \quad u = x^3$
 $du = 3x^2 dx$

$$= \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C = \frac{1}{3} e^{x^3} + C$$

Ex Area under $f(x) = e^{-3x}$ on $[0, 1]$

$$A = \int_0^1 e^{-3x} dx = -\frac{1}{3} \int_0^1 e^u du = -\frac{1}{3} [e^{-3} - 1]$$